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Effect of Natural and Artificial Light on Fe(III) Organic Complexes Photolysis: Case of Fe(III)-Malonate and Fe(III)-Malate

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Abstract:

The photolysis of iron(III) organic complexes in the presence of natural and simulated light was a concern for the role it plays within the environment in terms of producing hydroxyl radicals able to degrade organic products. In the present work, the study focused on two iron(III) organic complexes. The characterization of these complexes by spectrometry allowed to show their stability in environment with a stability constant equal to $\text{Log } \beta = 13.13$ for the complex Fe(III)-malonate, and $\text{Log } \beta = 20.85$ for the Fe(III)-malate complex. Photolysis of these complexes was carried out using a lamp at 365 nm at different pHs, and followed by several assay Fe(II), H_2O_2 and $\bullet\text{OH}$. The complexes present different photo-reactivities, in the case of Fe(III)-malate, the photolysis is favored by the Fenton process and generates hydroxyl radicals, contrarily to Fe(III)-malonate photolysis which generates an oxidizing $\bullet\text{CH}_2\text{COOH}$ radical without formation of H_2O_2 nor of $\bullet\text{OH}$. The presence of oxygen and the addition of isopropanol and chloroform show different effects on the complex photolysis. Complexes mineralization at natural irradiation was also studied using chemical oxygen demand (COD).

Keywords: Fe(III)-malate, Fe(III)-malonate, photolysis, sunlight

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1 Introduction

In the recent decades, carboxylic acids have received significant attention as being among the most common dissolved organic compounds present in natural environment (Perdue and Gjessing 1990; Thurman 1985). They are also considered as one of the dominant classes of organic compounds found in the atmosphere in several phases (Talbot et al. 1996; 1996a; 1995; Talbot et al. 1997a). They are found in rainwater (Guiang, Krupa, and Pratti 1984; Kumar et al. 1993; 1996; Khare et al. 1997), snow and ice (Gunz and Hoffmann 1990), on aerosol particles (Grosjean 1989; Kawamura and Ikushima 1994; Limbeck and Puxbaum 1999) and in the gas phase (Talbot et al. 1990; 1989; 1988; Kumar et al. 1996).

In the presence of ferric ions, these acids may form stable complexes with high molar absorptivities, and their absorption spectra shifted towards the visible range, which might make their use in sunlight possible as inducer in photodegradation process.

Many studies have recognized that the photoexcitation of Fe(III)-carboxylate complexes can result in the formation of both Fe(II) by the ligand-to-metal charge transfer (LMCT) reactions and H_2O_2 through the reduction of O_2 by photoexcited polycarboxylate (Faust and Zepp 1993; Fukushima and Tatsumi 1999; Southworth and Voelker 2003). As a consequence, $\bullet\text{OH}$ could be formed by oxidizing Fe(II) with H_2O_2 namely Fenton reaction, which can oxidize most of organic molecules quickly and non-selectively owing to its high oxidation potential ($E_0 = +2.80\text{V}$ versus NHE) and mineralize them into carbon dioxide and water (Liu et al. 2006). The reaction with H_2O_2 and Fe(II) can also generate Fe(III) thus, the process of Fenton can take place.

Fe(III)-polycarboxylate complexes have a photochemical activity depending on the considered complex. It is showed that the nature of the ligand is significantly involved on the complex photolysis (Abida 2005).

The effectiveness of the photooxydoreduction process of Fe(III)-polycarboxylate complexes depends on the quantum yields of Fe(II) formation.

The quantum yields for the photo-production of Fe(II) vary with the nature of the carboxylate ligand and pH value (Abrahamson, Rezvani, and Brushmiller 1994). Indeed, complexes with ligands holding a higher oxygen to carbon ratio tend to exhibit better photoreduction ability. Moreover, ligands containing OH, keto or diol

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functional groups in α -position exhibit higher quantum yields than unsubstituted carboxylates (Weller, Horn, and Herrmann 2013).

Among these complexes, the study was focused on Fe(III)-malonate and Fe(III)-malate, which can form stable complexes with different speciation. For Fe(III)-malonate, there is no formation of polynuclear or protonated species (Salvadó et al. 1989), but Fe(III)-malate may form dimer and trimer species (Timberlake 1964).

The photochemistry of these complexes is different, despite the fact that they undergo under irradiation both a ligand-to-metal charge transfer. However, the photolysis of Fe(III)-malonate complexes is assumed to generate an oxidizing $\cdot\text{CH}_2\text{COOH}$ radical. This species re-oxidizes the photogenerated Fe(II), which makes the quantum yield of Fe(II) very low (0.022), without formation of H_2O_2 and $\bullet\text{OH}$, as reported by Xiao (Xiao et al. 2014). A study detailing the mechanism of action of Fe(III)-malonate was suggested by Z. Wang (Wang et al. 2010).

Furthermore, Fe(III)-malate complexes can undergo a charge transfer at two different sites: the carboxylate group or the deprotonated hydroxyl group. C. Weller reported the different mechanisms of the complex photolysis without providing any details (Weller, Horn, and Herrmann 2013).

The literature reported very few articles on the use of these complexes in pollutant photo-degradation (Xiao et al. 2014).

The present work aims to study the photochemical behavior of these complexes, at concentrations close to those of the environment and with different proportions. The synthesis of Fe(III)-malate and Fe(III)-malonate complexes was initiated on the basis of the molar ratio method described by L. Wang (Wang 2008).

In the experimental conditions, the study on the characterization of these complexes reveals that their stoichiometry is different in this case involving a different behavior.

The influence of the pH, oxygen, and irradiation wavelength in photochemical process was investigated. For the mechanistic study, the hydroxyl and superoxide radicals were followed in order to give an orientation to the reaction. The environmental fate of these complexes was also tested under solar radiation since there is no bibliography on this work. The kinetics of photolysis was evaluated as well as the mineralization.

2 Experimental section

2.1 Chemicals

All reagents were of the purest commercially available grade and used without further purification. The solutions were prepared using ultrapure water. Ferric sulfate ($\text{Fe}_2(\text{SO}_4)_3$) was purchased from Riedel-de Haën. Malic and malonic acids (99 %) were provided by Biochem. Chloroform (99 %), sodium hydroxide (98 %), benzene (99.7 %), sodium acetate (99 %) and sulfuric acid (96 %) were supplied by Prolabo. 1,10-phenanthroline (> 99 %), TiCl_4 (98 %) and Phenol (~ 90 %) by Flucka. Isopropanol (99.8 %) was purchased from Scharlau.

2.2 Synthesis of organic iron complexes

The complex solutions were prepared by mixing ferric sulfate and organic acid solutions with a ratio Fe(III)/L (1/4) for Fe(III)-malonate and (1/2) for Fe(III)-malate according to the molar ratio method.

2.3 Irradiation experiments

2.3.1 Artificial irradiation

The irradiation experiments were performed in a Pyrex cylindrical reactor with a double-envelope with about 3 cm diameter with a cooling water jacket placed in an elliptical stainless steel chamber. The photo-reactor was exposed to the radiation sources composed of one low pressure mercury UV lamp Philips HPW 125 type (UV-A, $\lambda_{\text{max}} = 365 \text{ nm}$). The radiation intensity at this wavelength ($I = 1.97 \text{ mW.cm}^{-2}$) was measured using a UV radiometer type VLX-3W. The reaction temperature was kept at $20 \pm 1^\circ\text{C}$ by cycling water. The solution (usually 100 ml) was magnetically stirred during irradiation to insure homogeneity.

2.3.2 Natural irradiation

Solar irradiations were carried out in Constantine, Algeria (latitude $36^{\circ} 20' N$, longitude $6^{\circ} 37' E$) during June 2016 in sunny days. Experiments were conducted in a cylindrical Pyrex reactors placed vertically exposed to sunlight. Light intensity was measured using a Vilber VLX-3W, The average of solar irradiance measured at 365 nm was $2 \text{ mW} \cdot \text{cm}^{-2}$.

2.4 Analyses

The HPLC system consisted of a Shimadzu High Performance equipped with a photodiode detector (Shimadzu SPD-20A) and a Supelco HC-C18 column ($5 \mu\text{m}$, $250 \text{ mm} \times 4.6 \text{ mm}$). All the spectrophotometric experiments were carried out by a Thermo scientific spectrophotometer controlled by software « Thermo INSIGHT ».

Fe(II) concentrations were measured using the complexometric method with 1,10-phenanthroline (Seraghni et al. 2012), the molar coefficient at 510 nm is equal to $11,725 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$. The hydrogen peroxide concentration was also measured using the complexometric method with titanium tetrachloride (TiCl_4). This method is based on the formation of a yellow complex resulting from the reaction of H_2O_2 with Ti(IV) in the presence of sulphuric acid. The molar absorption coefficient at 410 nm is equal to $737 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$ (Kribéche et al. 2016). Scavenging $\bullet\text{OH}$ by high concentrations of benzene (7 mM) was used to determine the $\bullet\text{OH}$ radical formation, the high reactivity of benzene with $\bullet\text{OH}$ is $k = 8 \times 10^9 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ (Mameri et al. 2016), it was thought that $\bullet\text{OH}$ mediated oxidation of benzene would form phenol with a nearly 100 % yield (Faust and Allen 1993; Joseph, Varghese, and Aravindakumar 2001; Wang et al. 2006).

3 Results and discussion

3.1 Spectral properties of the complexes

The synthesis of complexes was carried out in the laboratory according to the protocol described in Section 2.4. In a first step, the characterization by spectrometry was useful for the follow-up of the complexes formed. The UV-visible absorption spectra of Fe(III), Fe(III)-malonate and Fe(III)-malate complexes are shown in Figure 1. The absorption spectrum of Fe(III) 0.3 mM in aqueous solution shows a strong absorption band at 292 nm. At the same time, addition of the ligands results in a change of the spectra by the appearance of new bands characterizing the formed complexes. Fe(III)-malate complex exhibited a stronger absorbance than Fe(III)-malonate in the visible range which affects its photochemical behaviour.

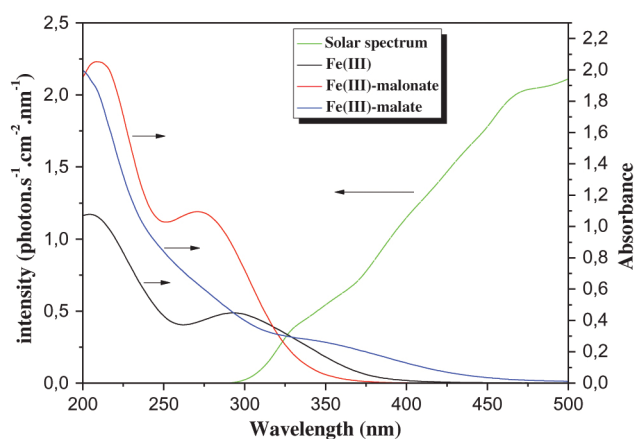


Figure 1: UV-Visible absorption spectra. Fe(III) 0.3 Mm; Fe(III)-malonate (0.3 mM:1.2 mM); Fe(III)-malate (0.3 mM:0.6 mM) complexes.

3.1.1 Determination of the stability constant

The determination of stability constant related to each complex was performed using the saturation method based on the spectrophotometric analysis (Seraghni et al. 2012). The concentration of metal ions (C_M) in aqueous solution was kept constant, and the concentration of the relative acid (C_L) was increased in the experiment.

The stability constant as well as the stoichiometry were deduced from the following equation (Seraghni et al. 2012):

$$\text{Log} \left(\frac{A_i}{A_{\max} - A_i} \right) = n \text{Log} C_L + \text{Log} \beta$$

Where:

A_i is absorbance corresponding to the growing part of the saturation curve.

$A_{\max}$ is absorbance corresponding to the level of the saturation curve.

B is constant of stability.

C_L is ligand concentration for each A_i .

n is (C_L/C_M) ratio.

Table 1 summarizes the Stoichiometry and stability constant of the complexes.

Table 1: Stoichiometry and stability constant of Fe(III)-malonate and Fe(III)-malate complexes.

Complex	Stoichiometry	Log β (our work)	Log β [reference]
Fe(III)-malonate	2: 1	13.13	13.29 (Salvadó et al. 1989)
Fe(III)-malate	2: 1	20.85	15.21 (Vukosav, Tomisic, and Mlakar 2010)

3.2 Complex speciation

The pH effect on the speciation of Fe(III)-malate and Fe(III)-malonate complexes was performed given the dependence of this factor on the photochemical process.

In order to determine the pKa value of the complex, various pH values were proceeded with.

3.2.1 Fe(III)-malate complex

Figure 2 shows the spectral evolution of Fe(III)-malate complex as a function of pH. Variations in the spectra were observed in the pH range of 2.5 to 6.9, with the appearance of two isobestic points at 333 and 355 nm. However, above 6.9 pH value, spectra remained virtually unchanged.

The absorbance at 280 nm and 310 nm increased rapidly with increasing pH, corresponding to the formation of the dimers and trimers as reported by Timberlake (Timberlake 1964).

The pKa value evaluated through the variation of absorbance at 334 nm was 3.5 (Figure 2).

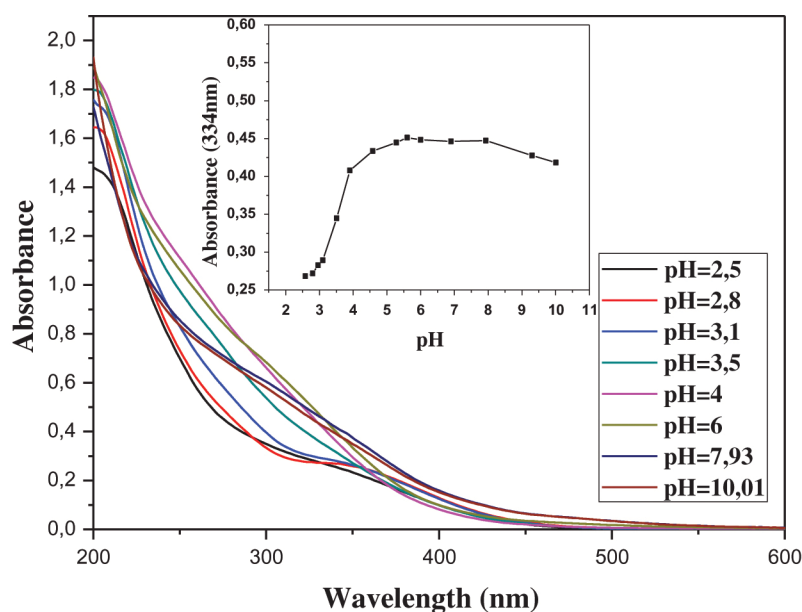


Figure 2: Spectral evolution of Fe(III)-malate (0.3 mM/0.6 mM) at different pH values. Insert displays absorbance as a function of pH.

3.2.2 Fe(III)-malonate complex

On the other hand, the UV-visible absorption spectra of the solutions containing Fe(III)-malonate (0.3 mM, 1.2 mM) change according to the increase in pH value between 2.8 and 6.5 (Figure 3).

It is to be mentioned that starting at pH = 4.5, an absorption band begins to appear in the visible domain and simultaneously, the UV band deforms and tends to disappear. The appearance of an isobestic point at 309 nm means that a transformation occurs.

Indeed, at pH > 5.5, a precipitation of ferric hydroxide was observed due to Fe(III)-malonate decomplexation (Salvadó et al. 1989). The pKa of the complex is 5.9 (Figure 3).

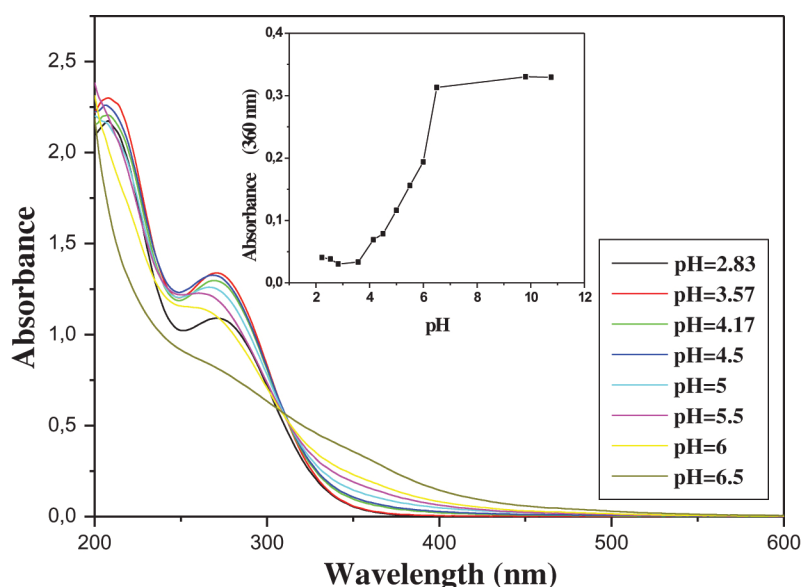


Figure 3: Spectral evolution of Fe(III)-malonate (0.3 mM/1.2 mM) at different pH values. Insert displays absorbance as a function of pH.

3.2.3 Stability of Fe(III) organic complexes

While studying the stability of Fe(III)-malonate and Fe(III)-malate, solutions stored in the dark and at room temperature for one week showed no evolution. Hence, the stock solutions of the complexes were refrigerated at 6 °C for an extended period.

3.3 Photochemical properties of Fe(III)-malonate and Fe(III)-malate complexes

The literature reported that Fe(III)-malonate and Fe(III)-malate complexes absorbed differently in the UV-visible region depending on the ligand type (Rodríguez et al. 2009). This would affect the photolysis kinetics of the complexes and the generated radicals.

The photochemical activity of the two organic complexes was tested by means of various parameters.

3.3.1 Fe(III)-malate photolysis

3.3.1.1 Spectrophotometric study

Photolysis of Fe(III)-malate complex (0.3 mM/0.6 mM, pH = 3) in aqueous solution was monitored using spectrophotometry under irradiation at 365 nm. The UV-Visible absorbance decreased strongly within the first 30 minutes. Moreover, the appearance of a new band at 283 nm confirms the photo-transformation of the complex (Figure 4).

The disappearance kinetics in aerated solution showed that 90 % of the complex disappeared after 30 min. irradiation. However, in deaerated solution, the disappearance of Fe(III)-malate complex was noticed to be very fast from the beginning of reaction. About 90 % of the complex disappeared after 10 minutes of irradiation (Figure 4).

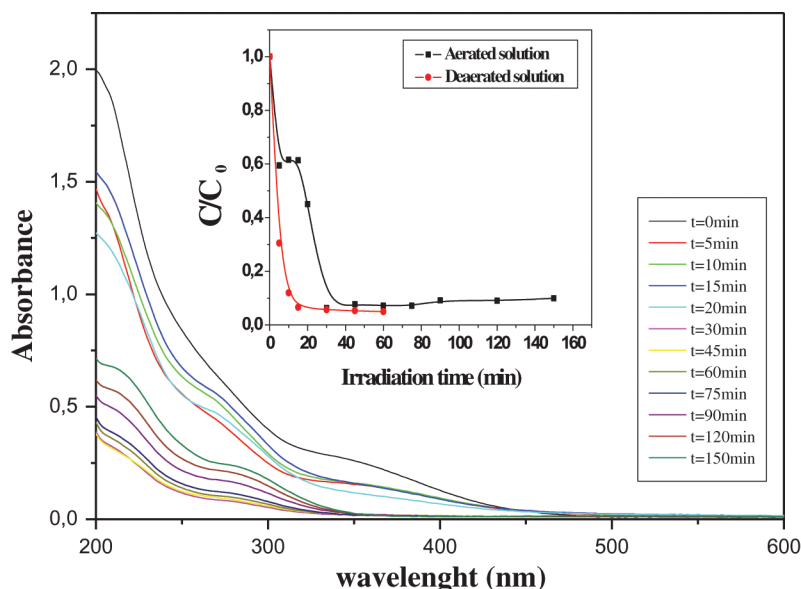
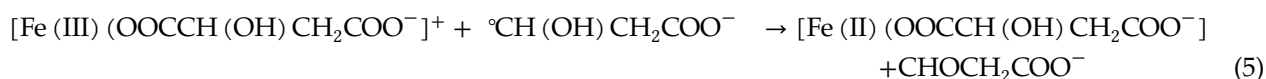
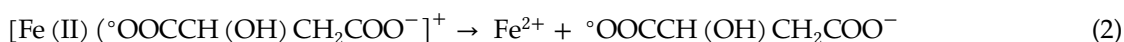
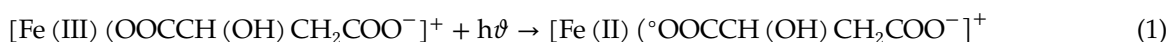


Figure 4: Spectral evolution of Fe(III)-malate complex (0.3 mM/0.6 mM, pH = 3) during the irradiation at 365 nm Inset displays disappearance kinetics of Fe(III)-malate complex in presence and absence of oxygen.

Some authors attributed this fact to the ligand structure. Weller reported that Fe(III)-malate is able to undergo a charge transfer (LMCT) at two different sites: the carboxylate group or the deprotonated hydroxyl group (Weller, Horn, and Herrmann 2013).

The absence of oxygen favoured the LMCT at the C1 carboxylate group (reactions [1] – [5]) where the radical formed after decarboxylation step attacked the parent complex according to the reaction [5] giving rise to an Fe(II)-ligand complex. The low stability constant of this complex reported by Vucosav (Vukosav, Tomisic, and Mlakar 2010) confirms the complex disappearance.



3.3.1.2 Formation of Fe(II) and H₂O₂

The formation of Fe(II) and H₂O₂ generated from the photolysis of the iron (III) organic complexes was monitored according to the protocol described in Section 2.4.

The Fe(II) formation kinetics is in perfect correlation with the complex photolysis reported. The photo-generation is fast from the beginning of the irradiation where the maximal concentration obtained after 30 min. was 1.7×10^{-4} M (Figure 5).

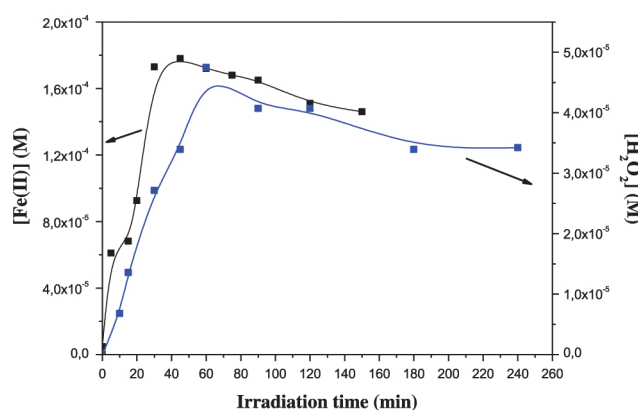


Figure 5: Fe(II) and H₂O₂ formation up on irradiation at 365 nm of the complex Fe(III)-malate (0.3 mM/0.6 mM, pH = 3).

H₂O₂ formation kinetics showed a similarity with that of Fe(II) wherein the concentration plateau (4.75×10^{-5} M) is reached after 60 minutes.

3.3.1.3 Effect of pH

The photolysis of Fe(III)-malate complex was studied in aqueous solution at various pH values, since it directly affects the complex speciation and therefore, influences significantly the photochemical process, because species have different reactivities linked to the photogeneration of hydroxyl radicals. Indeed, the maximal formation of $\bullet\text{OH}$ is obtained at pH = 3, indicating that the most active species are Fe₂L₂ and Fe₂L₃ (see formation $\bullet\text{OH}$ Section 3.4). The results summarized in Table 2, illustrate this fact.

Table 2: Fe(III)-malate disappearance and Fe(II) formation at different pH (0.3 mM/0.6 mM) at 365 nm.

pH	3.04	4.36	5.06	7.22
Complex disappearance after 30 min (%)	94	59	39	3
[Fe(II)] _{max} (10 ⁻⁴ M)	1.7	1.3	0.8	0.06

3.3.2 Photolysis of Fe(III)-malonate

3.3.2.1 Spectrophotometric study

In the same way, the photolysis at 365 nm of Fe(III)-malonate (0.3 mM/1.2 mM) was carried out in an aqueous solution and free pH. Spectral modification was observed during irradiation as shown in Figure 6. The characteristic band at 270 nm decreases with the appearance of an isobestic point at 330 nm. Moreover, photolysis kinetics are slow compared to Fe(III)-malate, 90 % of the complex disappeared after 240 min in an aerated solution as revealed by the Figure 6. The low absorbance of the complex at this wavelength explains this fact.

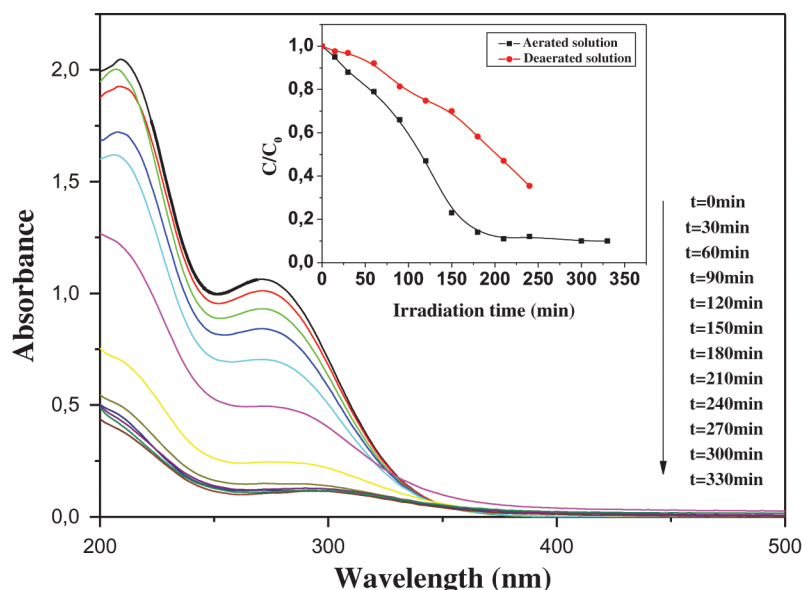


Figure 6: Spectral evolution during the irradiation at 365 nm of the complex Fe(III)-malonate (0.3 mM/1.2 mM, pH = 2.8).

The deficiency in oxygen slowed down the Fe(III)-malonate photolysis. This inhibition, although not total, is attributed to the nature of the organic ligand. The lack of oxygen disfavours the peroxy radical proposed by Wang on the mechanism diagram (Figure 7), it also promotes a low path of the complex decomposition named quenching mechanism.

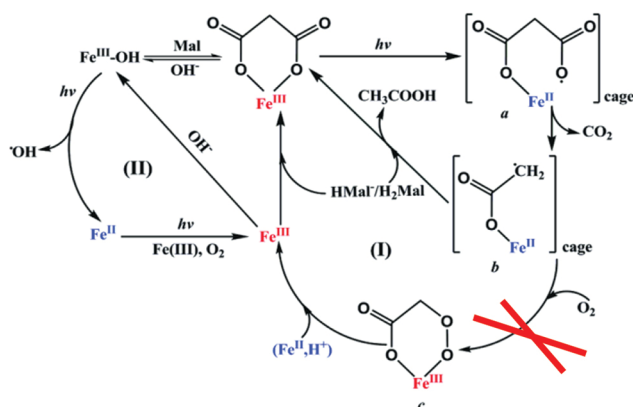


Figure 7: Fe(III)-malonate photolysis mechanism proposed by Z. Wang (Wang et al. 2010).

Insert displays complex Fe(III)-malonate kinetics disappearance in presence and absence of oxygen.

3.3.2.2 Formation of Fe(II) and H₂O₂

The Fe(II) formation kinetics in this case is distinguished by the presence of an induction phase where a low concentration was observed. However, for longer times (5 hours), the kinetics rise and reach a maximum value almost equal to that obtained in the case of Fe(III)-malate (1.6×10^{-4} M) (Figure 8).

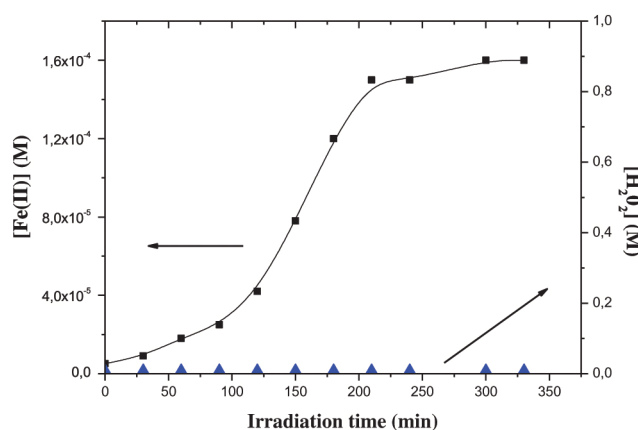
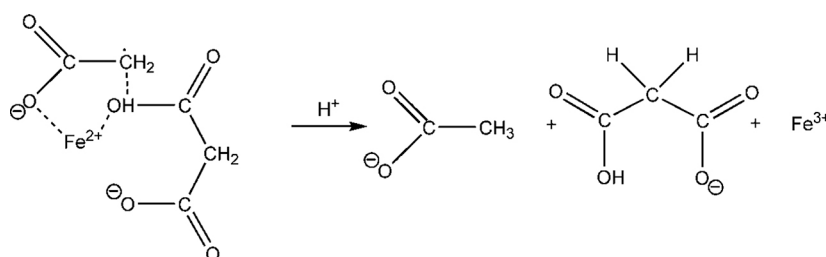


Figure 8: Formation of Fe(II) and H₂O₂ upon the irradiation of the Fe(III)-malonate complex (0.3 mM/1.2 mM, pH = 2.8).

This may be attributed to the quenching mechanism proposed by Z. Wang in the beginning of the photolysis (Wang et al. 2010). The radical complex formed reacts with free malonate ligand according to the reaction:



The rapid formation of Fe(II) observed in the second phase is attributed to Fe(OH)²⁺ species, which becomes dominant once the malonate ligand lows down.

The Fenton process described by the authors in organic iron (III) complex photolysis was verified by monitoring the formation of hydrogen peroxide and no peroxide formation was shown in this case. This result is in perfect correlation with that reported by Xiao (Xiao et al. 2014).

3.3.2.3 Effect of pH

The irradiation of Fe(III)-malonate complex at different pH values also showed strong dependence to the complex speciation. As the pH value increases, the photolysis slows down as illustrated in Table 3.

Table 3: Fe(III)-malonate disappearance and formation of Fe(II).

pH	2.8	3.6	4.5
Complex disappearance after 240 min (%)	90	53	20
[Fe(II)] _{max} (10 ⁻⁴ M)	1.6	0.22	0.09

This may be attributed to the distribution of complex species and their reactivity where Fe(Mal)₂⁻ is the most active species according to the speciation diagram (Salvadó et al. 1989). The kinetics of Fe(II) formation was also optimised at pH = 2.8 giving evidence to the involvement of the speciation in the photochemical process.

3.3.3 Mechanistic study

3.3.3.1 Involvement of radicals on photolysis

In order to give evidence for the involvement of radicals in complex photolysis, isopropyl alcohol 1–5 % (v/v) used as radical scavenger was added to both Fe(III)-malate and Fe(III)-malonate solutions.

In the first 15 minutes, the inhibition of Fe(III)-malate photolysis highlighted the involvement of radical species in the process. However, beyond this time the disappearance kinetics is not dependent on isopropanol content (Figure 9(a)).

On the other hand the irradiation of Fe(III)-malonate complex in the presence of 5 % of isopropanol indicated that the photolysis was completely inhibited. This inhibition may be explained by the deactivation of the excited state of the complex via isopropanol addition. This contrast showed a dependence of the ligand on the photolysis pathway which remains not elucidated (Figure 9(b)).

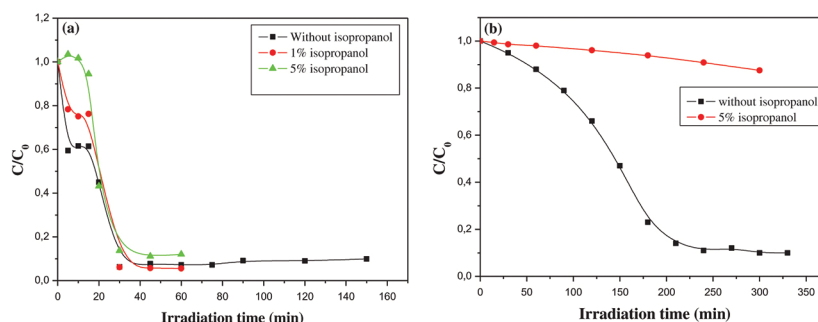


Figure 9: Effect of isopropanol in Complexes photolysis. Fe(III)-malate (0.3 mM/0.6 mM, pH = 3), (b) Fe(III)-malonate (0.3 mM/1.2 mM, pH = 2.8).

3.3.3.2 Effect of chloroform

The formation of $O_2^{\bullet-}$ radical plays a determining role in the generation of hydroxyl radicals during the photo-degradation process by producing H_2O_2 in Fenton reaction. Several authors proved the $O_2^{\bullet-}$ radical contribution in the photolysis process of organic iron complexes (Seraghni 2016).

Experiments with 1 and 5 % chloroform used as $O_2^{\bullet-}$ trap (Figure 10) showed that chloroform affects the photolysis of the Fe(III)-malate in both stages where the complex degradation kinetics is partially inhibited. Thus, the partial contribution of superoxide ion in photolysis was demonstrated.

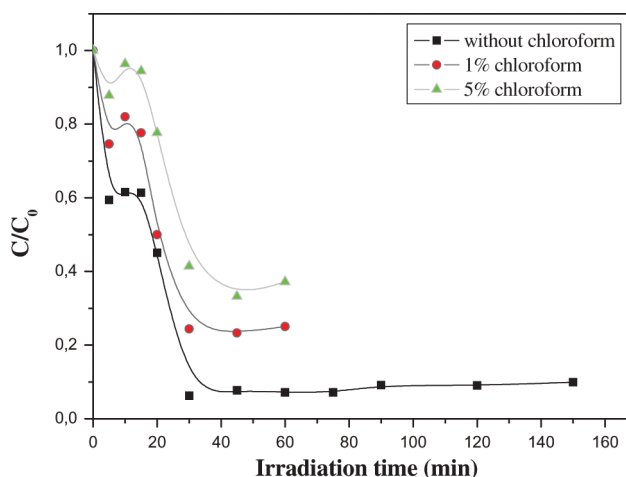


Figure 10: Photolysis Fe(III)-malate complex (0.3 mM/0.6 mM, pH = 3) in presence of chloroform.

At the same time, the followed formation of Fe(II) in these conditions showed that the addition of isopropanol or chloroform to Fe(III)-malate solution has no effect on its photogeneration. Nevertheless, in the case of Fe(III)-malonate, the amount formed in the plateau was lower than the detection limit (Table 4).

Table 4: Fe(II) formation during the complex photolysis in the presence of isopropanol and chloroform.

Isopropanol proportion		0%	1 %	5 %
Fe(II) concentration in the plateau (10^{-4} M)	Fe(III)-malate	1.73	1.72	1.71
	Fe(III)-malonate	1.6	/	0.04
Chloroform proportion		0 %	1 %	5 %
Fe(II) concentration (Fe(III)-malate) (10^{-4} M)		1.73	1.8	1.81

3.4 Determination of hydroxyl radicals

In general, the photolysis of organic Fe(III) complexes leads to the formation of hydroxyl radicals via Fenton process. The formation of this radical was followed during the photolysis of Fe(III)-malate and Fe(III)-malonate complexes using the method described in Section 2.4.

The concentration of $\bullet\text{OH}$ trained during the photolysis of Fe(III)-malate complex is very important because Fenton process is favoured by the coexistence of iron (II) and peroxide. However, for Fe(III)-malonate complex, the low content of hydroxyl observed in this case is attributed to another occurring process different from Fenton process due to the deficit in H_2O_2 (Figure 11).

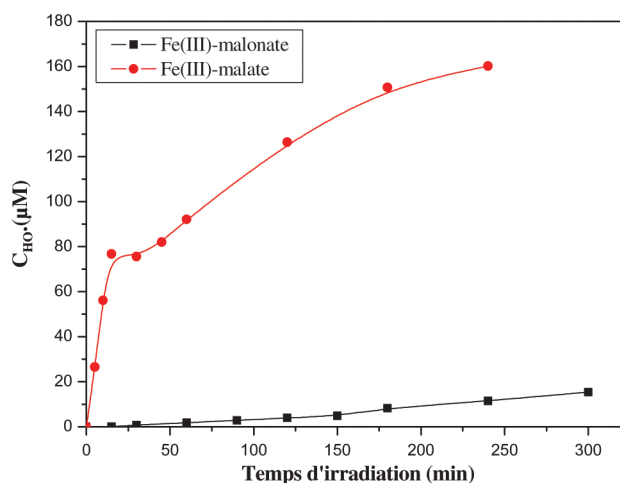


Figure 11: Hydroxyl dosage during the complex photolysis Fe(III)-malate (0.3 mM/0.6 mM, pH = 3); Fe(III)-malonate (0.3 mM/1.2 mM, pH = 2.8).

Moreover, the following of $\bullet\text{OH}$ formation during Fe(III)-malate photolysis at different pH values showed that the highest concentration was obtained at pH = 3. (Table 5) This fact helped conclude that the most active species are Fe_2L_2 and Fe_2L_3 .

Table 5: OH formation during the Fe(III)-malate photolysis at different pH values.

pH	3.04	4.36	5.06	7.22
[$\bullet\text{OH}$] (μM) after 180 min	150	33	16	2

3.5 Environmental significance

Photodegradation reactions in natural surface waters under solar light can take place when a substance absorbs solar radiation and thus, may become unstable and subsequently decompose and generates strong reactive species (such as superoxide anions, hydroxyl radicals or alkyl peroxy radicals), it can also react with organic compounds.

For better understanding of the organic complex behaviour in the surface waters, these complexes in solution state were exposed in natural light during a sunny day at Constantine (Algeria, Latitude $36^\circ 22' \text{N}$, longitude $6^\circ 40' \text{E}$).

By comparing the kinetics of disappearance at natural and artificial light (Figure 12), it appears that the rate of degradation is faster upon solar irradiation for both complexes because of the wide range of the solar spectrum.

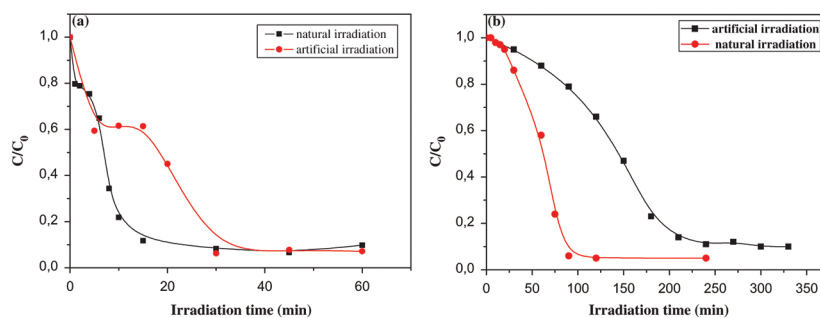


Figure 12: photolysis of complexes under artificial and natural irradiation Fe(III)-malate(0.3 mM/0.6 mM, pH = 3); (b) Fe(III)-malonate(0.3 mM/1.2 mM, pH = 2.8).

The Fe(II) formation kinetics is also faster under solar irradiation but reached the same artificial plateau for malate. However, the malonate generates a higher amount (1.9×10^{-4} M) (Figure 13). Its consumption by the Fenton reaction in the first case explains this result.

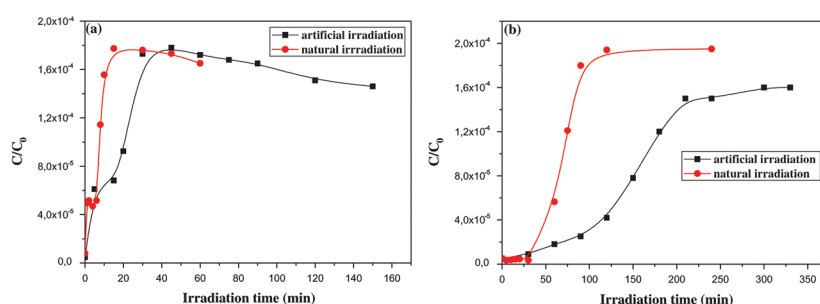


Figure 13: Fe(II) formation during photolysis of complexes under artificial and natural irradiation: Fe(III)-malate(0.3 mM/0.6 mM, pH = 3); (b) Fe(III)-malonate(0.3 mM/1.2 mM, pH = 2.8).

3.6 Mineralization

In order to extent to mineralization, COD served to follow the transformation to inorganic carbon, as is reported in experimental section. The results shows that after 5 h of natural irradiation, the mineralization of the ligand is not total for the two complexes, 74 % for the Fe(III)-malate and 50 % in the case of Fe(III)-malonate Figure 14. This is due to the refractory intermediates products photo-generated by photolysis. These intermediates needs more long time for malonate than malate, although malate contains more carbon.

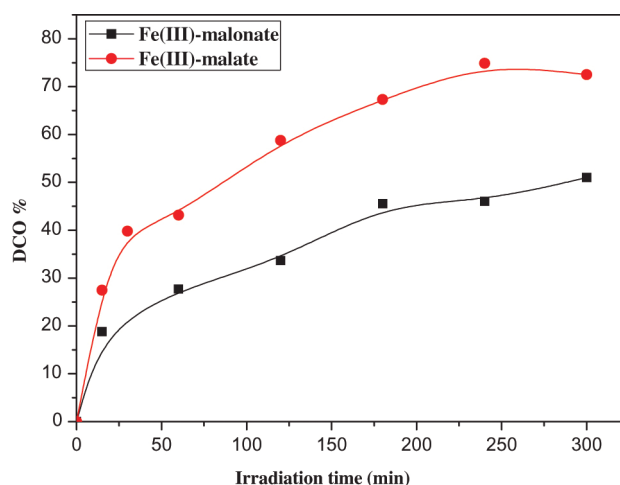


Figure 14: Mineralization of Fe(III)-malate (0.3 mM/0.6 mM, pH = 3) and Fe(III)-malonate (0.3 mM/1.2 mM, pH = 2.8) under natural irradiation.

4 Conclusion

In this paper, the photochemical reactivity of two iron organic complexes (Fe(III)-malate and Fe(III)-malonate) is investigated in aqueous solution at 365 nm. The synthesis and characterization of these complexes has been carried out within the laboratory. Their characterization by spectrophotometric method is preliminary studied. The saturation method allowed evaluating the constant of stability and stoichiometry (7.07×10^{20} for Fe(III)-malate and 1.34×10^{13} for Fe(III)-malonate, the stoichiometry is 2: 1 for both complexes).

Up on irradiation at 365 nm both complexes undergo photolysis whose kinetics depend organic ligand. This is attributed to the molar absorption coefficient of the complex. The pH medium, optimized in acid medium, affects the photochemical process. The importance of oxygen in the photolysis process was highlighted. Thus for Fe(III)-malonate the lack of O_2 inhibits the photolysis while for Fe(III)-malate it was accelerated. This is related to the radical formed, after the charge transfer ligand to metal, whose reactivity with oxygen is dissimilar. To prove the Fenton process established by different authors, we followed the formation of Fe(II) and H_2O_2 via the complex photolysis. The results let see that this hypothesis possesses some exception. In our experimental conditions, Fe(III)-malate has generate Fe(II), H_2O_2 and consequently $\bullet OH$. However, no formation of H_2O_2 and $\bullet OH$ during Fe(III)-malonate photolysis. All of these observations promote much interest in the understanding of the mechanisms implicate in complex phototransformation.

The study under solar irradiation showed that the photolysis of the two complexes is accelerated with a mineralization of 74 % for Fe(III)-malate and 50 % for Fe(III)-malonate after 5 h.

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